

Homework 1

- This is a modified version of a recent final exam given in ST 501.
 - You are allowed to discuss answers and generally collaborate on assignments in this class.
 - However, you are not allowed to copy answers/explanations. You should be writing things in your own words and writing out derivations yourself.
 - If you are ever questioning whether a collaboration is allowed, please just ask!
 - **Although there are points associated with each problem, only selected problems will be graded for correctness, the rest will be graded for completeness (and not worth the points as noted).** The points are just to show you how I would allocate this on an exam.
1. Briefly answer the following:
 - (a) What does the term distribution mean in non-technical terms? (3 pts)
 - (b) In non-technical terms, what is meant by the phrase 'sampling distribution of the sample mean'? (2 pts)
 - (c) Practically speaking, what does the Central Limit Theorem tell us (be specific about assumptions required and clear about what statistics it applies to!) and why is it so useful? (5 pts)
 - (d) In regards to statistical inference, why do we often make the assumption that a sample is 'iid'? (3 pts)

Again, please note: **Show all of your work and explain non-algebra steps** to receive full credit.

2. Suppose that $Y_i \stackrel{iid}{\sim} \text{exp}(\lambda)$.

(a) Derive the mean and variance of $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$. (3 and 5 pts, respectively)

(b) Give a detailed argument (state any theorems and assumptions) to show that a reasonable large-sample distribution for $\bar{Y}$ is a Normal distribution (and of course give the mean and variance). (6 pts)

(c) Give an equation in terms of a standard normal random variable's CDF, n , and λ that will approximate $P(\bar{Y} > 20)$. (3 pts)

- (d) (This question is part of the previous problem.) Find a large-sample distribution for the $\ln(\bar{Y})$ (don't derive the exact distribution here, give an approximating distribution). (6 pts)

- (e) (This question is part of the previous problem.) Consider $S^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$. Prove that $S^2 \xrightarrow{p} 1/\lambda^2$. Again, be sure to state theorems and results you use. (8 pts)

(f) (This question is part of the previous problem.) Give an argument as to why

$$Z = \frac{\bar{Y} - (1/\lambda)}{S/\sqrt{n}}$$

converges in distribution to a Normal distribution. Again, be sure to state theorems and results you use. (6 pts)

3. Suppose that $Y \sim \text{geom}(p)$ where Y represents the number of failures until the first success. That is, we have support $y=0,1,2,\dots$ (this differs from the formula sheet version). Information about this version of the geometric is given below.

$$p_Y(y) = p(1-p)^y \quad E(Y) = (1-p)/p \quad \text{Var}(Y) = (1-p)/p^2 \quad m_Y(t) = \frac{p}{1-(1-p)e^t}$$

Please note the geometric sum properties on the formula sheet.

- (a) Derive the MGF of Y given above. (6 pts)

- (b) Consider having an integer b . Show that $P(Y > b) = (1-p)^{b+1}$ (6 pts)

- (c) Now suppose we have a random sample, $Y_i \stackrel{iid}{\sim} \text{geom}(p)$. Show that $W = \sum_{i=1}^n Y_i \sim \text{NegativeBinomial}(n, p)$. (8 pts) With this version of the geometric, W represents the number of failures until the n^{th} success and has support $w = 0, 1, 2, \dots$. Information about this version of the negative binomial is given below.

$$p_W(t) = \binom{w+n-1}{w} (1-p)^w p^n \quad E(W) = r(1-p)/p \quad \text{Var}(W) = r(1-p)/p^2 \quad m_W(t) = \left(\frac{p}{1-(1-p)e^t} \right)^n$$

- (d) Suppose that $n \rightarrow \infty$ and $(1 - p) \rightarrow 0$ in such a way that $n(1 - p) \rightarrow \lambda$. Show that $W \xrightarrow{d} X \sim Poi(\lambda)$. Recall our common limit trick, which appears on the formula sheet. (8 pts)

4. Let $Y \sim \text{Beta}(\alpha, \beta)$.

(a) Derive the result $E(Y^r) = \frac{\Gamma(\alpha+r)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+r+\beta)}$. Assume $r > -\alpha$. (6 pts)

(b) Use the above result to find the variance of Y . (4 pts)

(c) **Two things:** Provide an equation that uses the PDF (no need to solve the equation) that corresponds to $P(0.2 < Y \leq 0.4)$ and draw a detailed picture of the PDF corresponding to this probability (pick any reasonable shape for the distribution of a Beta when drawing your picture). (6 pts)

(d) **Two things:** Give a detailed equation for how to find the 0.25 quantile of Y 's distribution (no need to simplify) using the PDF. Also, draw a detailed picture of this quantile in terms of the PDF. (6 pts)

Formula Sheet

Counting and Basic Probability

$$P_r^n = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!} \quad C_r^n = \binom{n}{r} = \frac{n!}{r!(n-r)!} \quad \binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \cdots n_k!}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad P(A) + P(A^C) = 1 \quad P(A \cap B) = P(A|B)P(B) \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A) = \sum_{i=1}^k P(B_i)P(A|B_i) \quad P(B_j|A) = \frac{P(B_j)P(A|B_j)}{P(A)} = \frac{P(B_j)P(A|B_j)}{\sum_{i=1}^k P(B_i)P(A|B_i)}$$

Random Variable Definitions and such

$$p_Y(y) = P(Y=y) \quad F_Y(y) = P(Y \leq y) \quad f_Y(y) = \frac{d}{dy} F_Y(y) \quad F_Y(y) = \int_{-\infty}^y f_Y(t) dt$$

pmf/pdf must have two properties: $f_Y(y) \geq 0$ or $p_Y(y) \geq 0$ and integral or sum over entire support must be 1.

$$E(g(Y)) = \begin{cases} \sum_y g(y)p_Y(y) & \text{if } Y \text{ is discrete} \\ \int_{-\infty}^{\infty} g(y)f_Y(y)dy & \text{if } Y \text{ is continuous} \end{cases} \quad P(a \leq Y \leq b) = \begin{cases} \sum_{y:a \leq y \leq b} p_Y(y) & \text{if } Y \text{ is discrete} \\ \int_a^b f_Y(y)dy & \text{if } Y \text{ is continuous} \end{cases}$$

$$\text{Mean: } \mu_Y = E(Y) \quad \text{Variance: } \sigma_Y^2 = \text{Var}(Y) = E[(Y - \mu_Y)^2] = E(Y^2) - \mu_Y^2$$

$$m_Y(t) = E(e^{tY}) \quad E(Y^n) = m_Y^{(n)}(0) = \frac{d^n}{dt^n} m_Y(t)|_{t=0}$$

$$E(a+bY) = a+bE(Y) \quad \text{Var}(a+bY) = b^2 \text{Var}(Y) \quad p^{\text{th}} \text{ quantile of } F_Y(y) \text{ is value } y_p \text{ s.t. } F_Y(y_p) = p$$

Random Math Things

$$\text{Binomial Theorem: } (x+z)^n = \sum_{y=0}^n \binom{n}{y} x^y z^{n-y} \quad \text{Maclaurin Series of } e^\lambda = \sum_{y=0}^{\infty} \frac{\lambda^y}{y!}$$

$$\text{Finite Sum of Geometric: } \sum_{y=0}^{n-1} ap^y = \frac{a(1-p^n)}{1-p} \quad \text{Sum of Geometric: } \sum_{y=0}^{\infty} ap^y = \frac{a}{1-p}$$

$$\text{Other geometric sum properties: } \sum_{y=1}^{\infty} yp^{y-1} = \frac{1}{(1-p)^2} \quad \sum_{y=0}^{\infty} \frac{p^{y+1}}{y+1} = -\ln(1-p)$$

$$\text{If } a_1, \dots, a_n \text{ converges to } a \text{ then } \left(1 + \frac{a_n}{n}\right)^n \rightarrow e^a$$

$$\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} e^{-t} dt \quad \Gamma(\alpha+1) = \alpha\Gamma(\alpha) \text{ for } \alpha > 0 \quad \Gamma(n+1) = n! \text{ for } n \text{ integer} \quad \Gamma(1/2) = \sqrt{\pi}$$

Discrete Distributions

Binomial: $p_Y(y) = \binom{n}{y} p^y (1-p)^{n-y}$ support: $y=0,1,2,\dots,n$

$$E(Y) = np \quad Var(Y) = np(1-p) \quad M_Y(t) = (pe^t + (1-p))^n$$

Geometric: $p_Y(y) = (1-p)^{y-1} p$ support $y=1,2,\dots$ For Integer $b, P(Y > b) = (1-p)^b$

$$E(Y) = 1/p \quad Var(Y) = (1-p)/p^2 \quad M_Y(t) = \frac{pe^t}{1-(1-p)e^t} \quad t < -\ln(1-p)$$

Neg Bin: $p_Y(y) = \binom{y-1}{r-1} p^r (1-p)^{y-r}$ support: $y=r, r+1, r+2, \dots$

$$E(Y) = r/p \quad Var(Y) = r \frac{(1-p)}{p^2} \quad M_Y(t) = \left(\frac{e^t p}{1-(1-p)e^t} \right)^r \quad t < -\ln(1-p)$$

Hypergeometric: $p_Y(y) = \frac{\binom{r}{y} \binom{n-r}{m-y}}{\binom{n}{m}}$ support: $y=0,1,2,\dots,m$

$$E(Y) = rm/n \quad Var(Y) = \frac{mr(n-r)(n-m)}{n^2(n-1)} \quad \text{No closed form MGF}$$

Poisson: $p_Y(y) = \frac{\lambda^y e^{-\lambda}}{y!}$ support: $y=0,1,2,\dots$

$$E(Y) = \lambda \quad Var(Y) = \lambda \quad M_Y(t) = e^{\lambda(e^t - 1)}$$

Continuous Distributions

Gamma: $f_Y(y) = \frac{\lambda^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\lambda y}$ support: $0 < y < \infty$

$$E(Y) = \alpha/\lambda \quad Var(Y) = \alpha/\lambda^2 \quad M_Y(t) = \left(\frac{1}{1-t/\lambda} \right)^\alpha, \quad t < \lambda$$

If $\alpha=1$ then $Y \sim \exp(\lambda)$. If $\alpha=p/2, \lambda=1/2$, then $Y \sim \chi_p^2$

Normal: $f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(y-\mu)^2}$ support: $-\infty < y < \infty$

$$E(Y) = \mu \quad Var(Y) = \sigma^2 \quad M_Y(t) = e^{\mu t + \sigma^2 t^2 / 2}$$

If $Z = \frac{Y-\mu}{\sigma}$ then $Z \sim N(0,1)$.

Beta: $f_Y(y) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} y^{\alpha-1} (1-y)^{\beta-1}$ support: $y \in [0,1]$

$$E(Y) = \frac{\alpha}{\alpha+\beta} \quad Var(Y) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} \quad M_Y(t) = \text{Not useful}$$

Uniform: $f_Y(y) = \frac{1}{\theta_2 - \theta_1}$ support: $y \in [\theta_1, \theta_2]$

$$E(Y) = \frac{\theta_1 + \theta_2}{2} \quad Var(Y) = \frac{(\theta_2 - \theta_1)^2}{12} \quad M_Y(t) = \frac{e^{\theta_2 t} - e^{\theta_1 t}}{(\theta_2 - \theta_1)t}$$

Transformation Theorems

If $X \sim N(\mu, \sigma^2)$ then $Y = aX + b \sim N(a\mu + b, a^2\sigma^2)$

If $X \sim f_X(x)$, a continuous RV, and $Y = g(X)$, where g is differentiable and strictly monotonic over the support of X , then $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$

Probability Integral Transform: $X \sim F_X(x)$, a continuous RV with strictly increasing cdf, $Y = F_X(x) \sim U(0,1)$

Let $U \sim U(0,1)$ and define $X = F^{-1}(U)$, then X has cdf F .

Joint Distributions

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y) \quad p_{X,Y}(x,y) = P(X=x, Y=y) \quad p_X(x) = \sum_y p_{X,Y}(x,y) \quad P((X,Y) \in A) = \sum_{(x,y) \in A} p_{X,Y}(x,y)$$

Multinomial: $p_{Y_1, \dots, Y_n}(y_1, \dots, y_n | p's) = \binom{m}{y_1, \dots, y_n} \prod_{i=1}^n p_i^{y_i}$ support: $y_i \geq 0, y_1 + \dots + y_n = m$

Marginally, $Y_i \sim \text{Bin}(m, p_i)$

$$F_{X,Y}(x,y) = \int_{-\infty}^y \int_{-\infty}^x f_{X,Y}(s,t) ds dt \quad \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y) = f_{X,Y}(x,y) \quad P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x,y) dy dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

If $(X,Y) \sim \text{BVN}(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$ then marginally, $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ and conditionally, $X|Y=y \sim N(\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y), \sigma_X^2(1 - \rho^2))$

Conditional Distributions

$$p_{X|Y}(x|Y=y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} \quad f_{X|Y}(x|Y=y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

If joint cdf/pdf/pmf is product of marginal cdf/pdf/pmf then $Y_1, Y_2, \dots, Y_n$ are independent.

X is independent of Y if and only if $f_{X,Y}(x,y) = g(x)h(y)$

Order Statistics

Distribution of j^{th} order statistic: $f_{Y_{(j)}}(y) = n \binom{n-1}{j-1} f_Y(y) [F_Y(y)]^{j-1} [1 - F_Y(y)]^{n-j}$

Joint Distribution of i^{th} and j^{th} order statistic:

$$f_{Y_{(i)}, Y_{(j)}}(u,v) = n f_Y(u) (n-1) f_Y(v) \binom{n-2}{i-1, j-i-1, n-j} [F_Y(u)]^{i-1} [F_Y(v) - F_Y(u)]^{j-i-1} [1 - F_Y(v)]^{n-j}$$

Expected Values of a function of multiple RVs

$$E(g(Y_1, \dots, Y_n)) = \begin{cases} \sum_{y_1} \dots \sum_{y_n} g(y_1, \dots, y_n) p_{Y_1, \dots, Y_n}(y_1, \dots, y_n) \\ \int_{y_1} \dots \int_{y_n} g(y_1, \dots, y_n) f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) dy_n \dots dy_1 \end{cases}$$

If X is independent of Y then $E(g(X)h(Y)) = E(g(X))E(h(Y))$

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - E(X)E(Y) \quad Corr(X, Y) = \rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

$$Var\left(a + \sum_{i=1}^n b_i X_i\right) = \sum_{i=1}^n \sum_{j=1}^n b_i b_j Cov(X_i, X_j) = \sum_{i=1}^n b_i^2 Var(X_i) + 2 \sum_{i < j} b_i b_j Cov(X_i, X_j)$$

$$E(g(X)|Y=y) = \begin{cases} \sum_{x|y} g(x) p_{X|Y}(x|y) \\ \int_{x|y} g(x) f_{X|Y}(x|y) dx \end{cases} \quad E(X) = E(E(X|Y)) \quad Var(X) = E(Var(X|Y)) + Var(E(X|Y))$$

Delta Method approx for $Y = g(X)$, (first order) $\mu_Y \approx g(\mu_X)$, $\sigma_Y^2 \approx [g'(\mu_X)]^2 \sigma_X^2$ (second order)
 $\mu_Y \approx g(\mu_X) + \frac{1}{2} g''(\mu_X) \sigma_X^2$

Limit Theorems etc.

$$Y_n \xrightarrow{p} Y \quad \text{if} \quad \lim_{n \rightarrow \infty} P(|Y_n - Y| \geq \epsilon) = 0 \quad \text{or} \quad \lim_{n \rightarrow \infty} P(|Y_n - Y| < \epsilon) = 1$$

$$Y_n \xrightarrow{d} Y \quad \text{if} \quad \lim_{n \rightarrow \infty} F_{Y_n}(y) = F_Y(y) \quad \text{or} \quad \lim_{n \rightarrow \infty} M_{Y_n}(y) = M_Y(y)$$

Convergence in probability implies convergence in distribution. The converse is true when you converge in distribution to a constant.

Markov's Inequality - If X is a nonnegative RV where $E|X| < \infty$ then for $t > 0$, $P(X \geq t) \leq E(X)/t$

Chebychev's Inequality - X a RV with μ, σ^2 , then for $t > 0$ $P(|X - \mu| \geq t) \leq \sigma^2/t^2$

WLLN - Y_n a sequence of independent RVs with $E(Y_i) = \mu$ then $\bar{Y}_n = \sum_{i=1}^n Y_i/n \xrightarrow{p} \mu$

Continuity Theorem - If Y_n converges to Y , g a continuous ftn, then $g(Y_n)$ converges to $g(Y)$

CLT - Y_i iid with $\sigma^2 < \infty$ then $\frac{\bar{Y}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$ or $\bar{Y}_n \sim N(\mu, \sigma^2/n)$ or $\sqrt{n}(\bar{Y}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$

Delta method normality - Y_n RVs where $Y_n \sim N(\theta_0, \sigma^2)$ then for a function g and value θ_0 where $g'(\theta_0)$ exists and is not 0 $g(Y_n) \sim N(g(\theta_0), [g'(\theta_0)]^2 \sigma^2)$.

Slutsky's Theorem - If $X_n \xrightarrow{d} X, Y_n \xrightarrow{p} a$ then $X_n Y_n \xrightarrow{d} aX$ and $X_n + Y_n \xrightarrow{d} X + a$